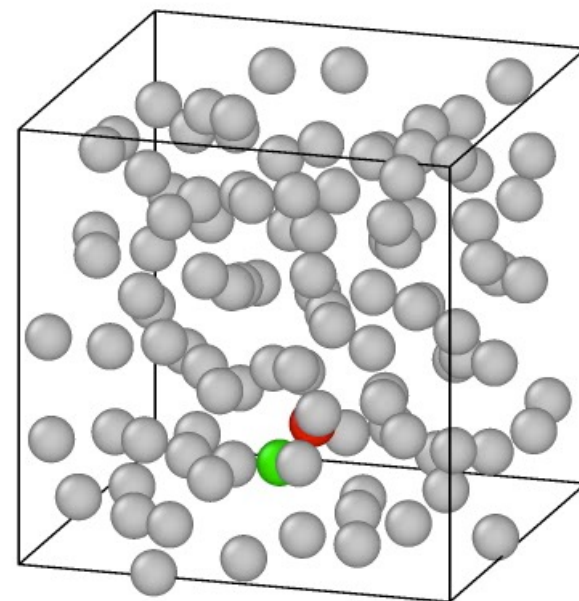


Determining melting curves without solid insight

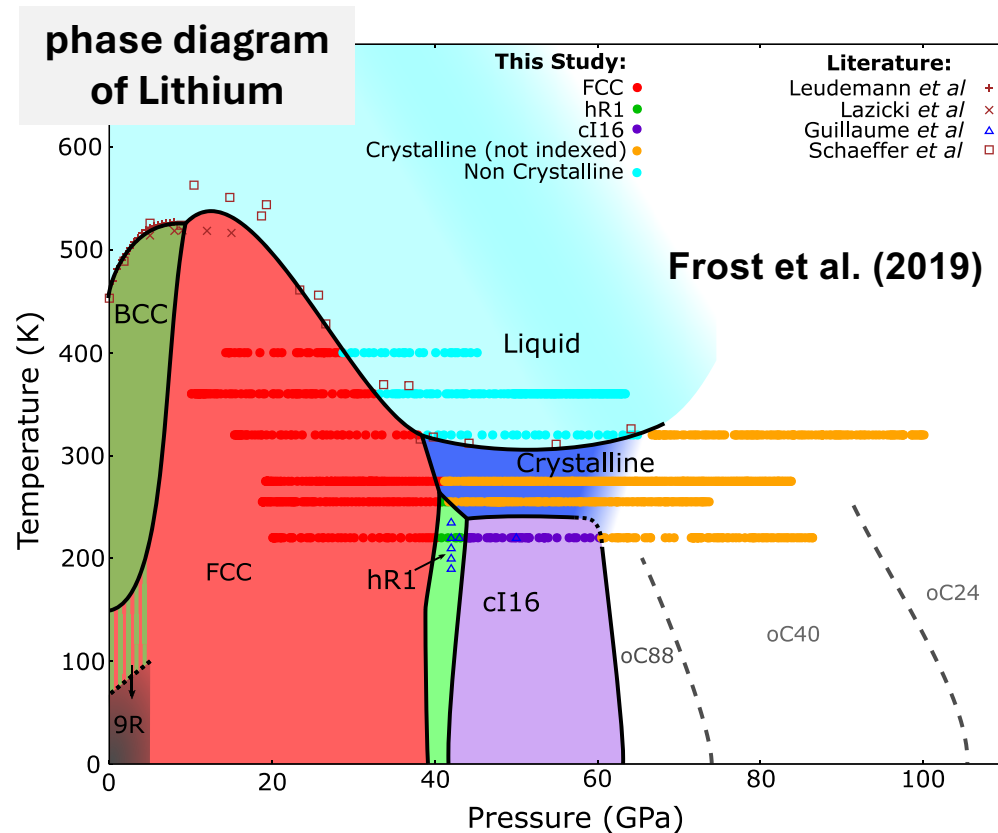
Jérôme Daligault

with Alfred Farris and Romain Perriot

Can the freezing point be predicted from
the liquid alone — with no knowledge of
the solid it turns into?



The continuing challenge of determining melting curves



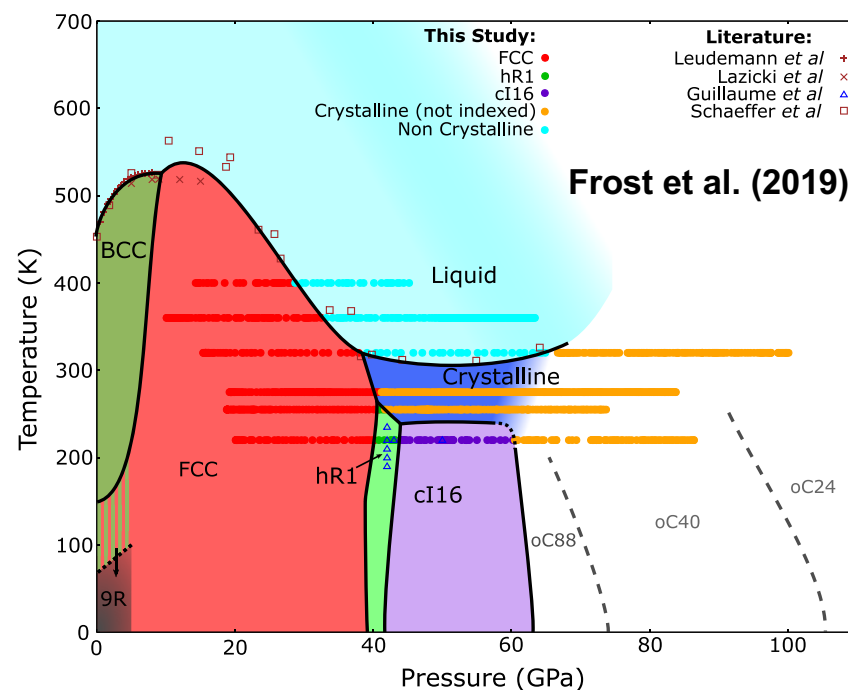
The melting curve is a bellwether for changes in bonding, structure and physical properties under pressure

Melting curves remain difficult to determine experimentally — especially under extreme conditions

Theoretical prediction is essential for interpreting and extending phase diagrams

Most Methods Depend on the Solid — a Nontrivial Prerequisite

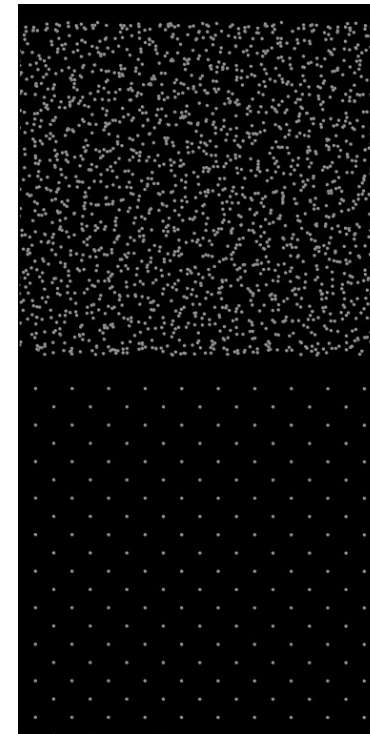
Method	Requires solid phase	Comment
Free energy	Yes	Exact in principle, expensive in practice
Two phase coexistence	Yes	Physically grounded, setup is delicate simulations are demanding
Lindemann	Yes	Empirical threshold on atomic displacements
Z method	Yes	Based on superheating the solid
Heat-until-melt	Yes	Superheating limit, gives upper bound



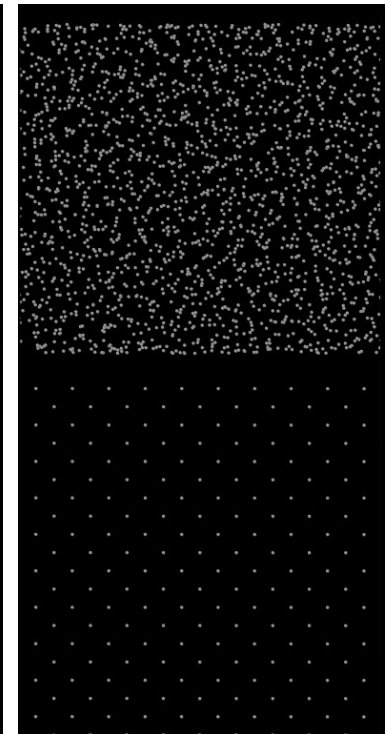
Most Methods Depend on the Solid — a Nontrivial Prerequisite

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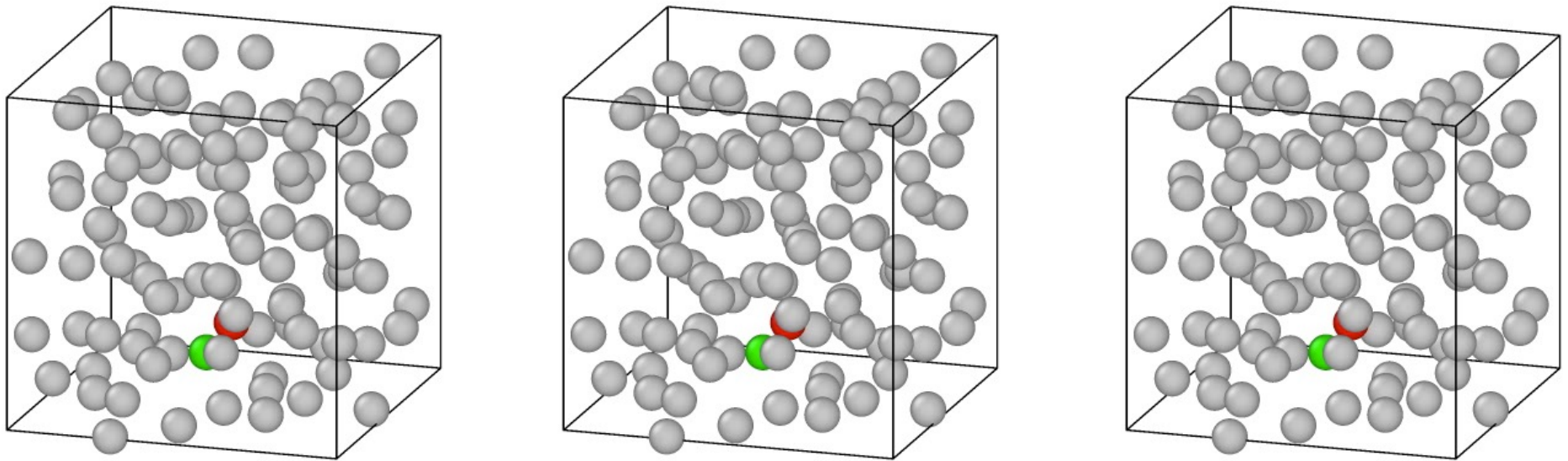
$T < T_{melt}$



$T > T_{melt}$



Can we do without the solid?



**Can we read the freezing point
from the liquid alone ?**

This talk:

- **A signal in the liquid**

A robust dynamical signature of the liquid–solid coexistence point in elemental liquids

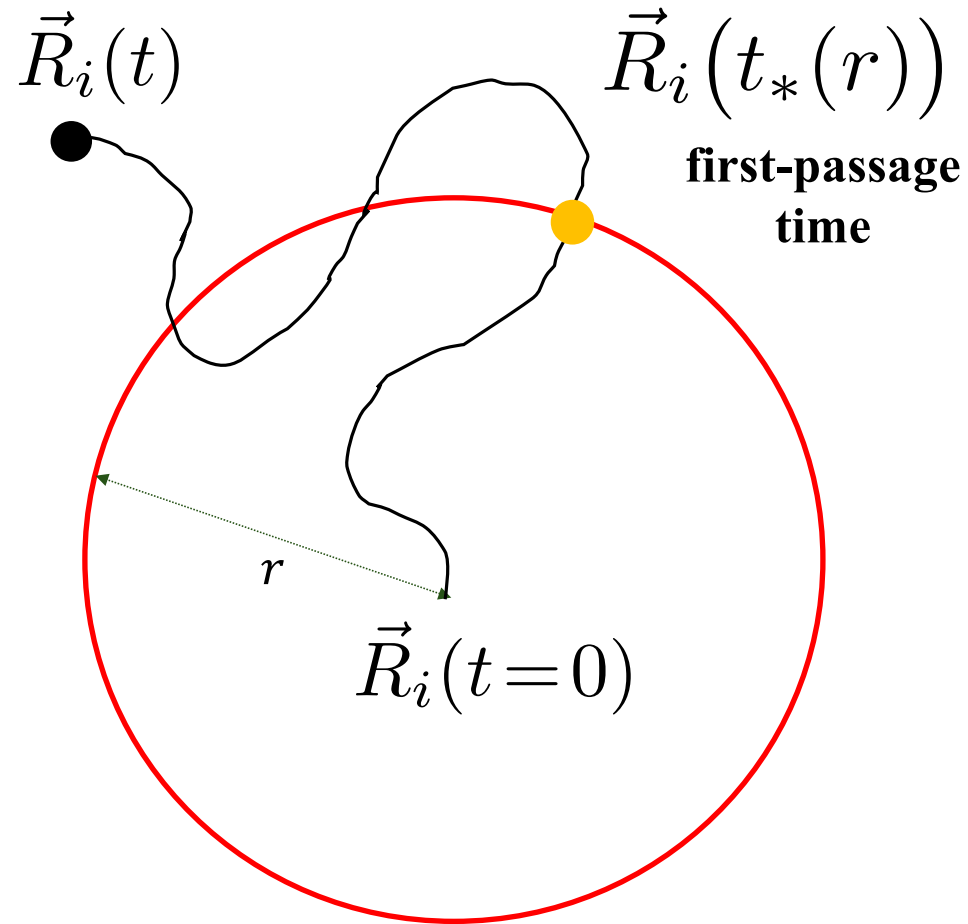
- **A method without the solid**

A practical way to locate melting curves using only liquid-phase simulations

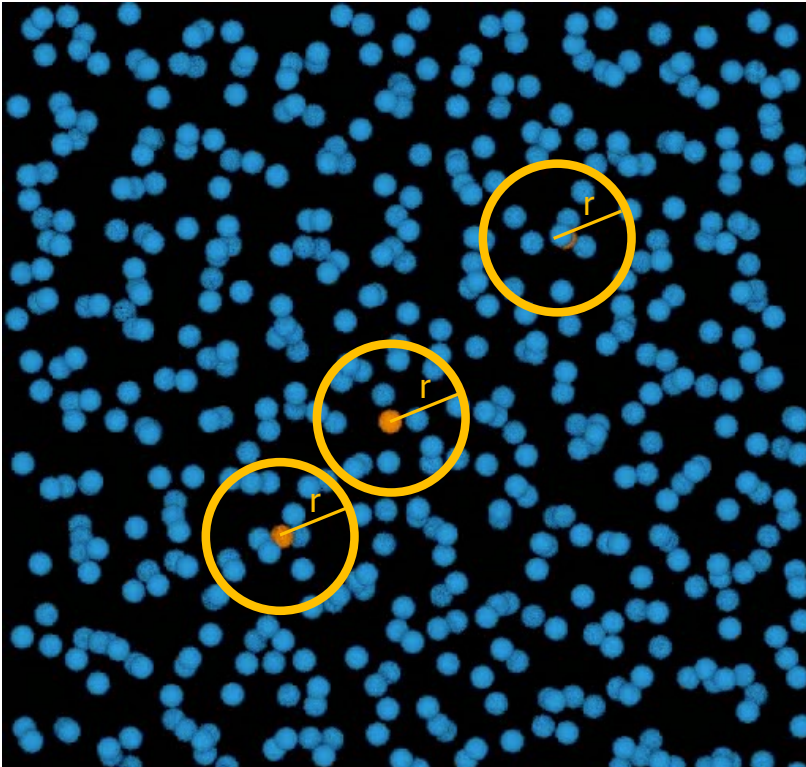
- **A dynamical view of thermodynamic coexistence**

Recasting coexistence as something encoded in atomic motion

A first-passage perspective on atomic motions



A first-passage perspective on atomic motions



mean first-passage time:

$$\tau(r) = \int_0^\infty dt \, t F_r(t)$$

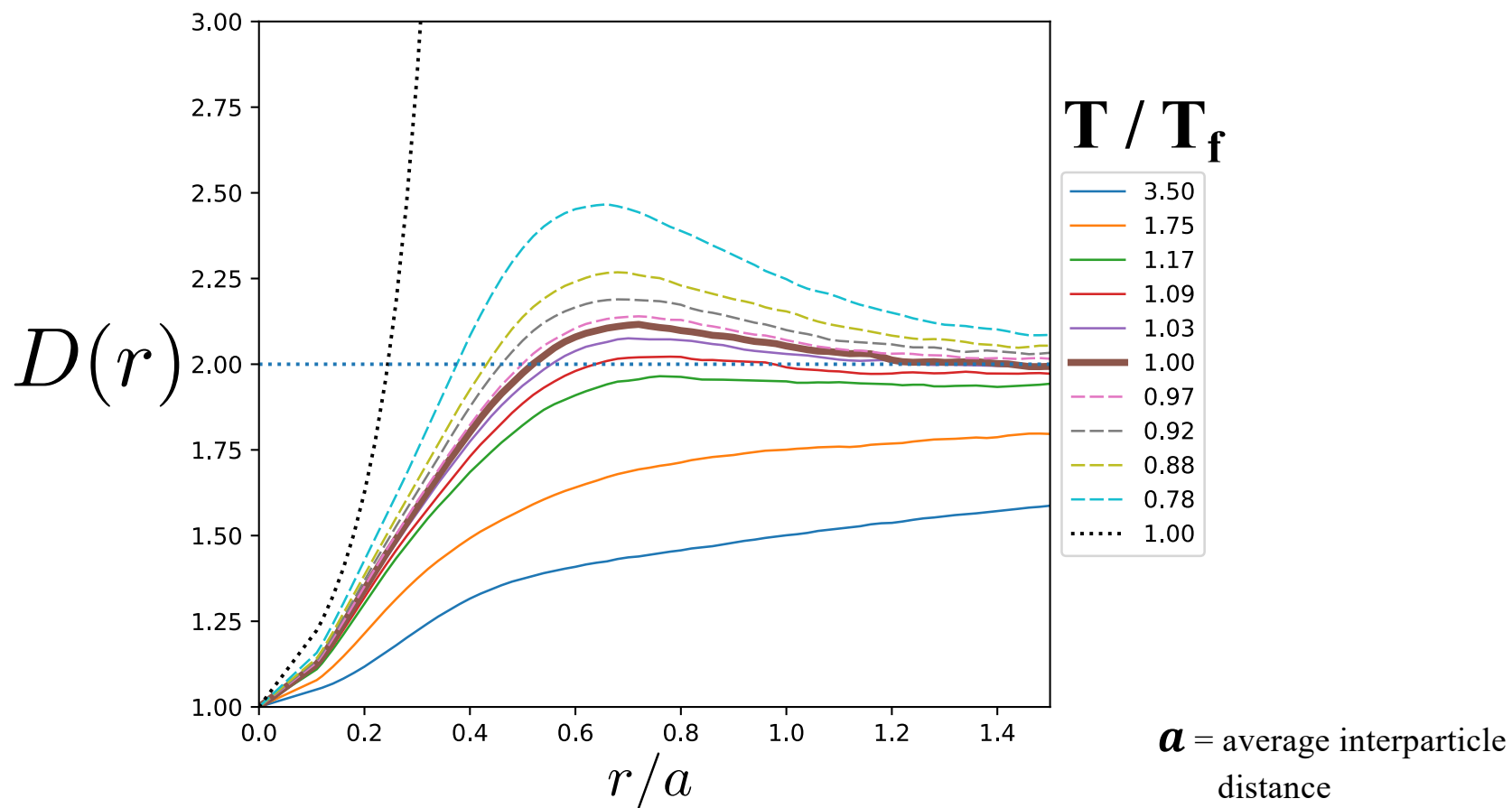
It is revealing to examine

$$D(r) = \frac{d \ln \tau(r)}{d \ln r}$$

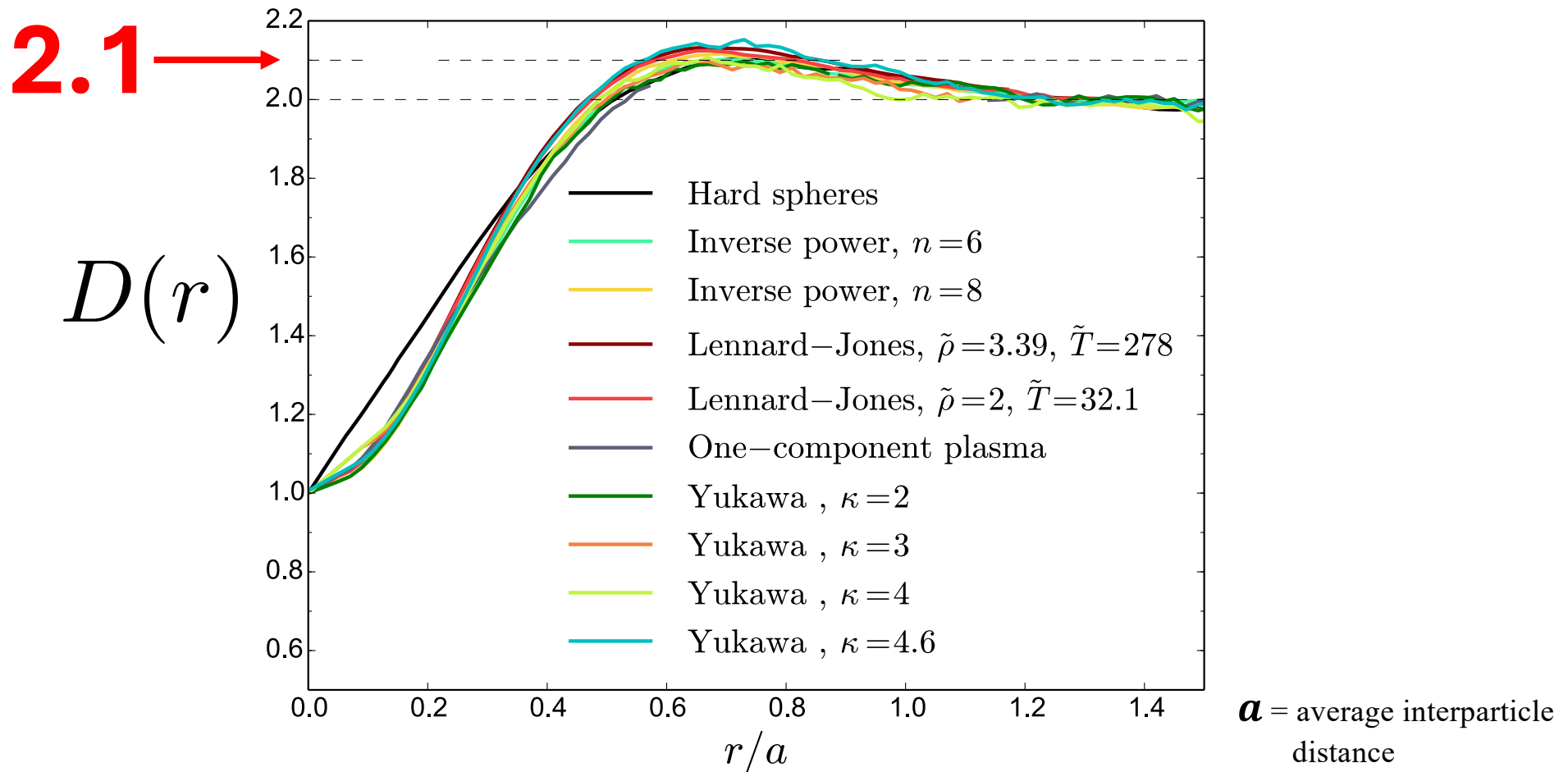
Basic motivations:

- $\tau(r) \propto r^d \Rightarrow D(r) = d$
- $D(r)$ is dimensionless
- In the fluid phase, $D(r)$ is expected to vary from 1 as $r \rightarrow 0$ (inertial regime) to 2 as $r \rightarrow \infty$ (diffusive limit)

$D(r)$ changes character near the freezing point

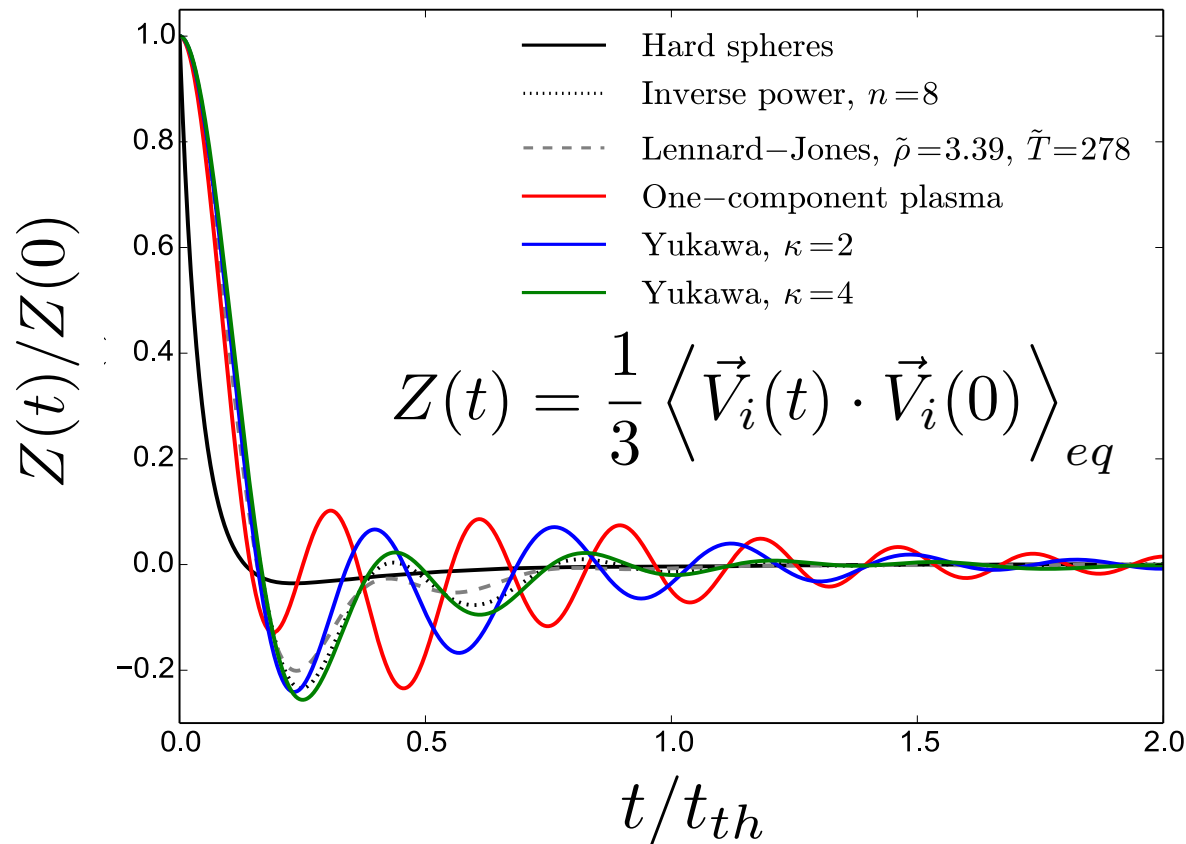


A signature of freezing point in $D(r)$



This is a surprising result

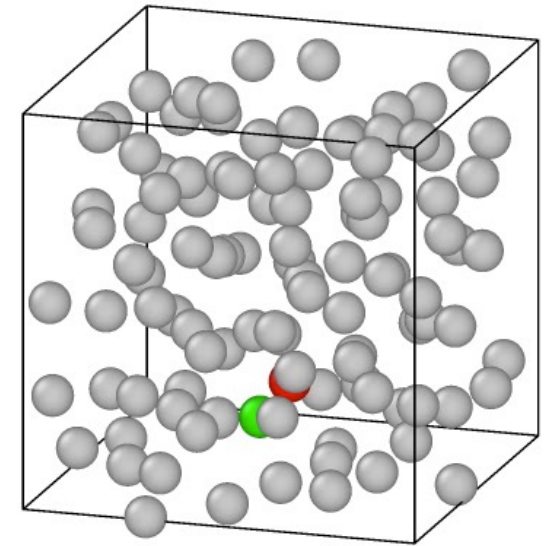
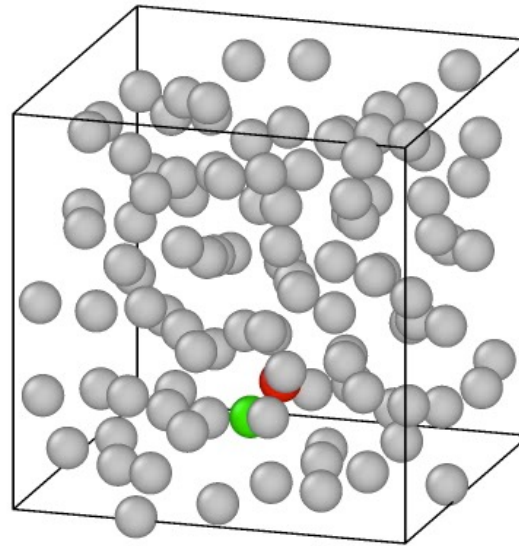
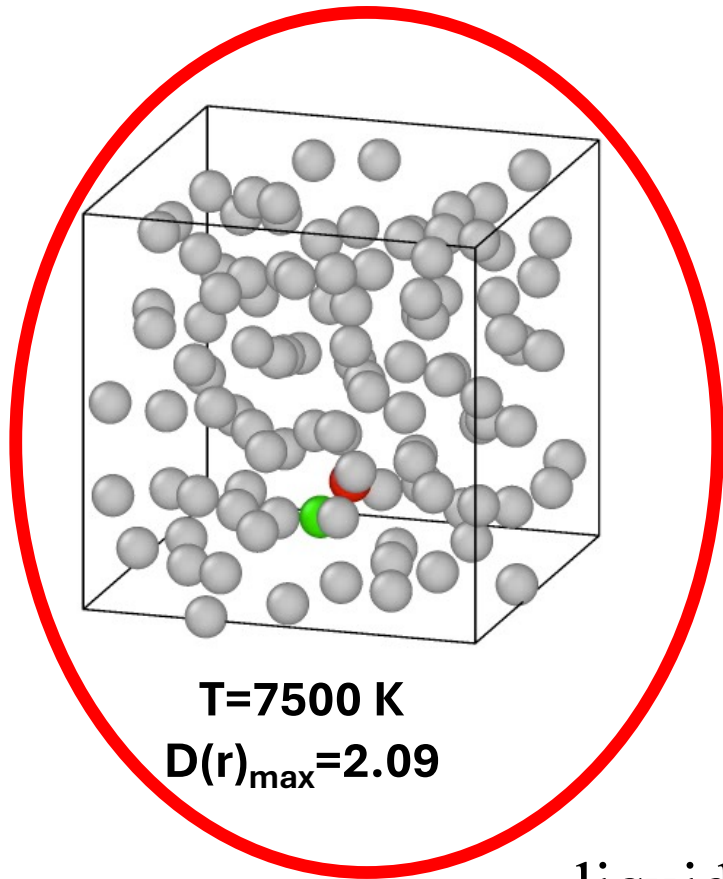
Traditional dynamical properties show system-dependent variation



Velocity autocorrelations do not scale similarly across systems at freezing point

$$t_{th} = a/v_{th} = a/\sqrt{3k_B T/m}$$

Answering the quiz: which liquid is at freezing?



liquid Aluminum @ 5.5 g/cm³

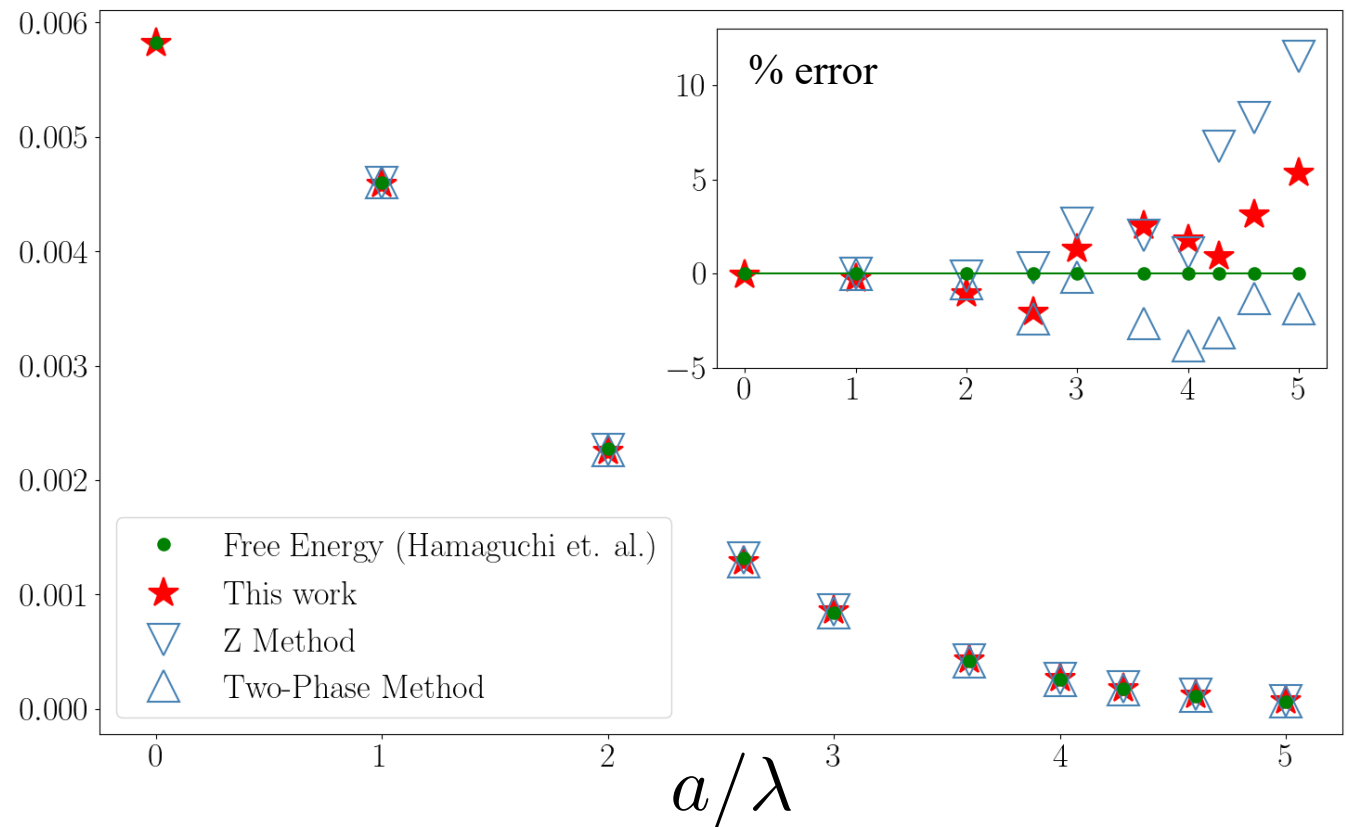
Making the heuristic a method:

Through benchmarking, sensitivity analysis, and optimization

melting
temperature T_m
(arb. units)

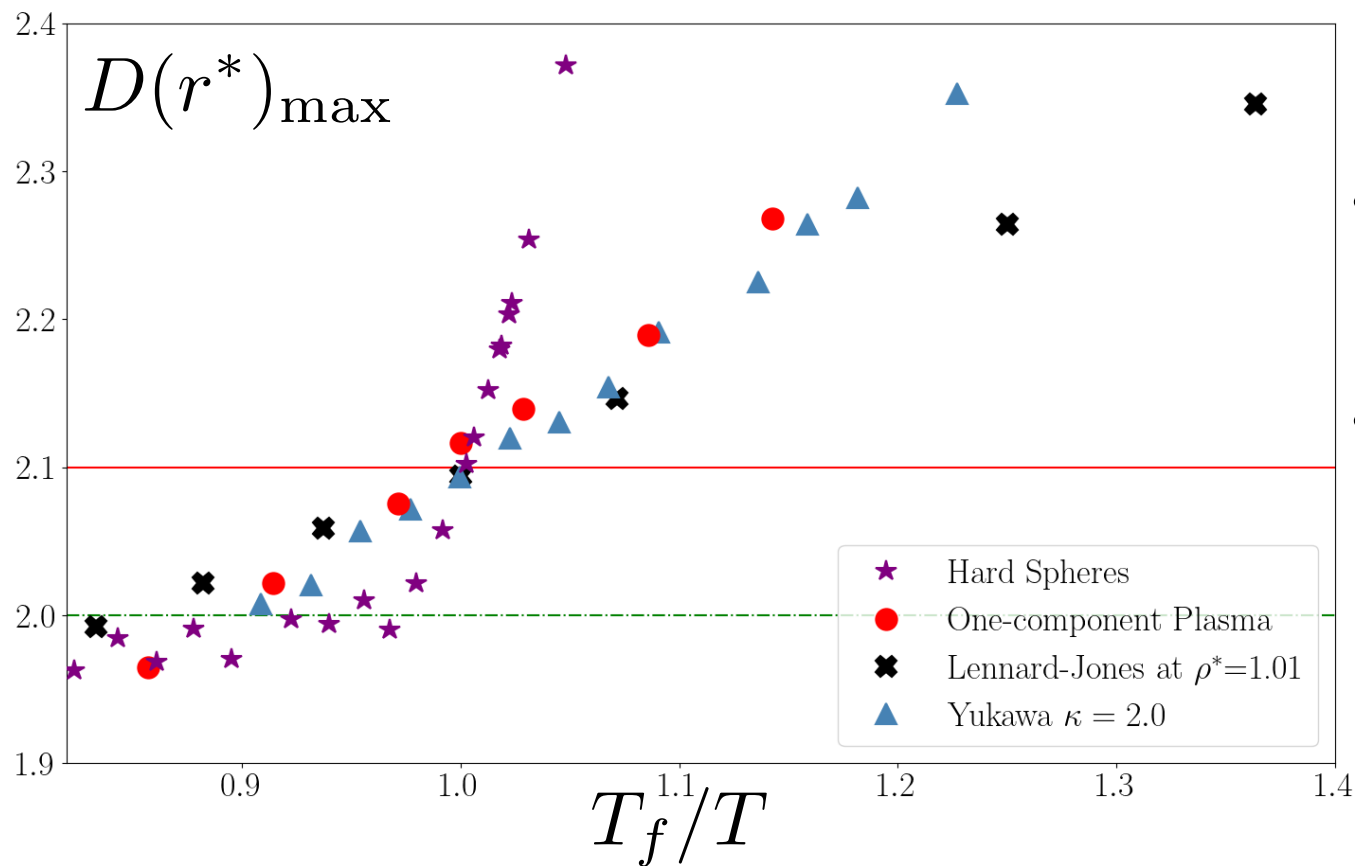
Yukawa model:

$$\phi(r) = \frac{q^2}{r} e^{-r/\lambda}$$



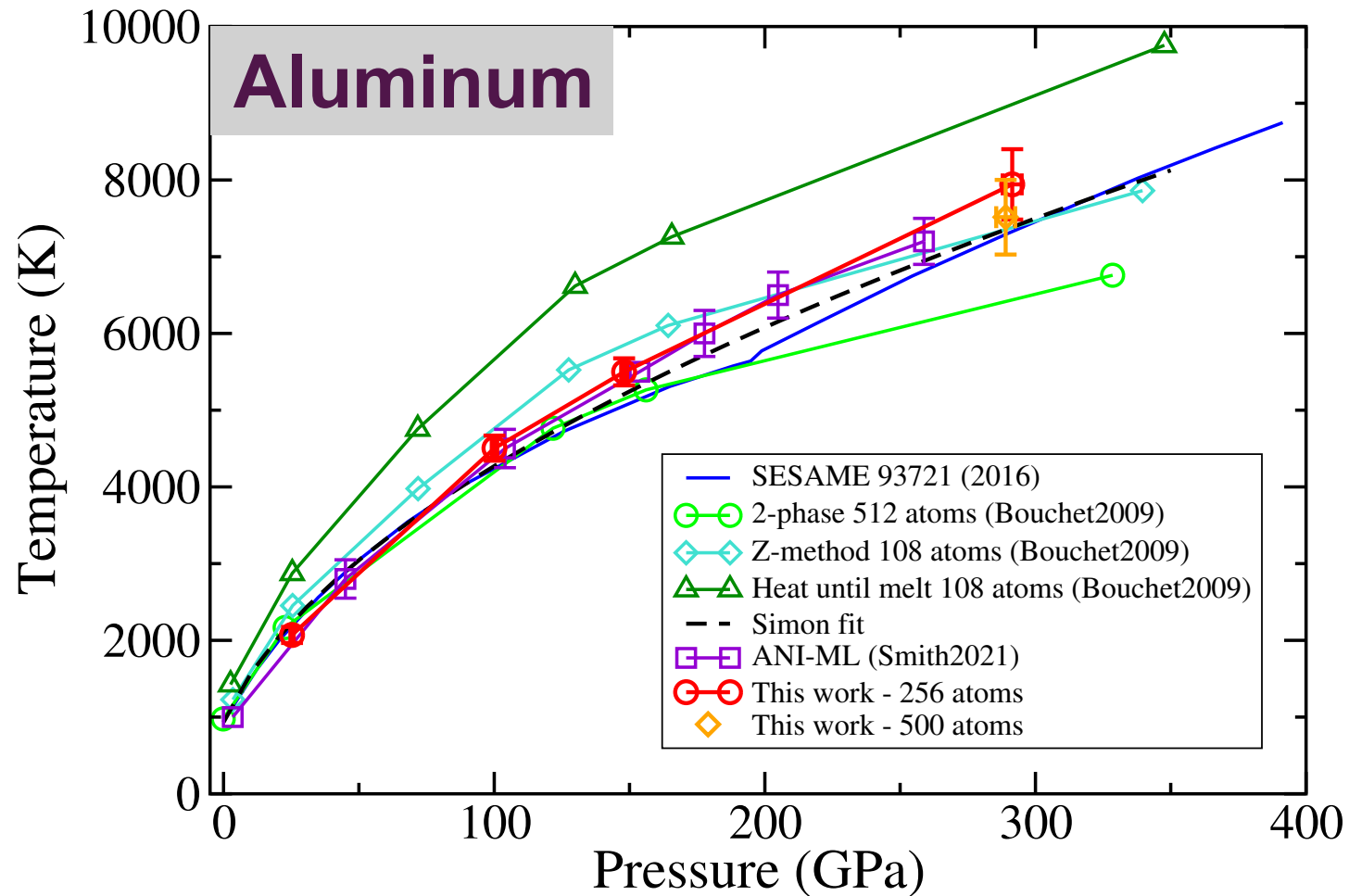
Making the heuristic a method:

Through benchmarking, sensitivity analysis, and practical optimization

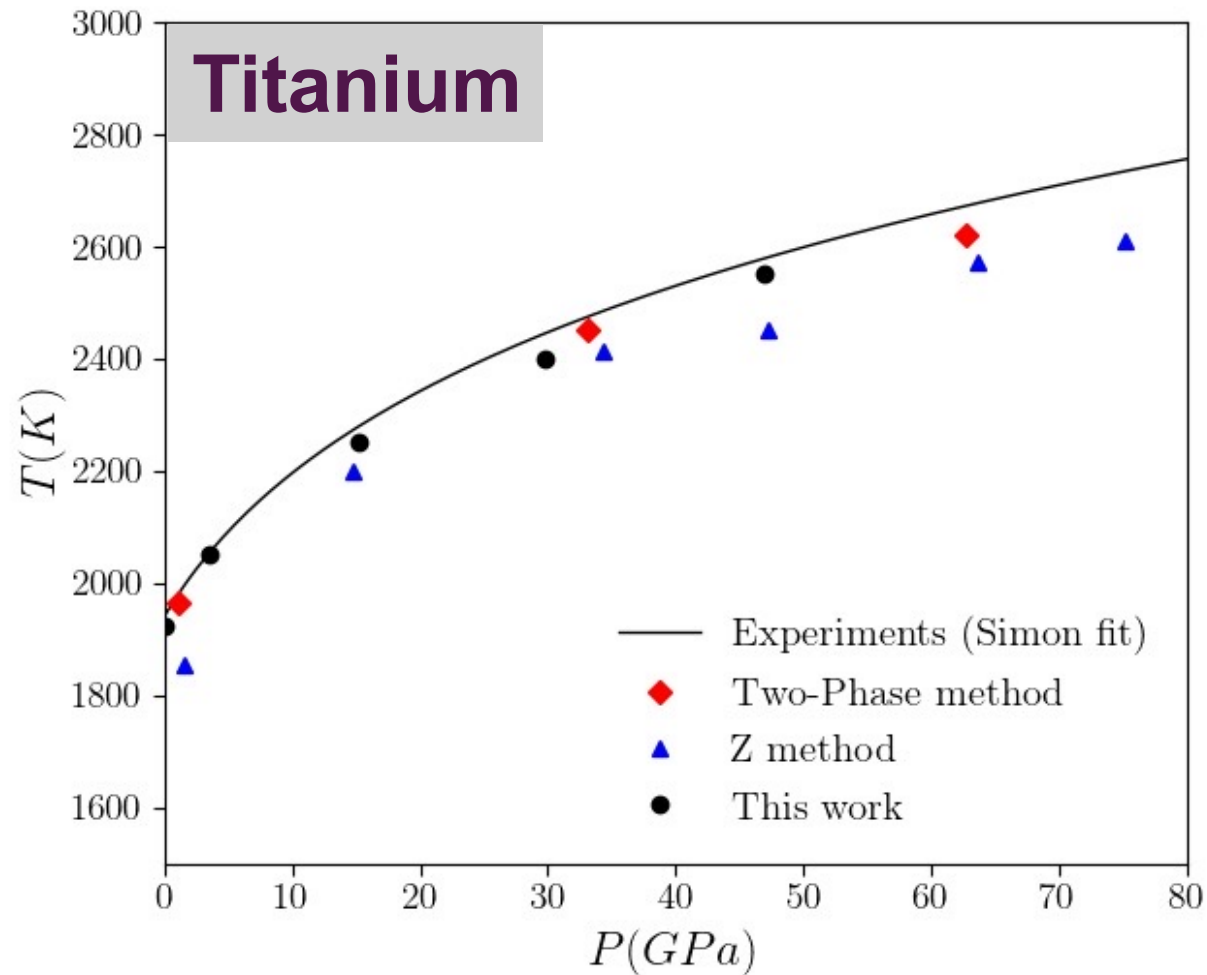


- The maximum value of $D(r)$ varies linearly from $2.0 < D(r) < 2.3$
- Interpolation greatly increases the efficiency of the method

From toy models to materials

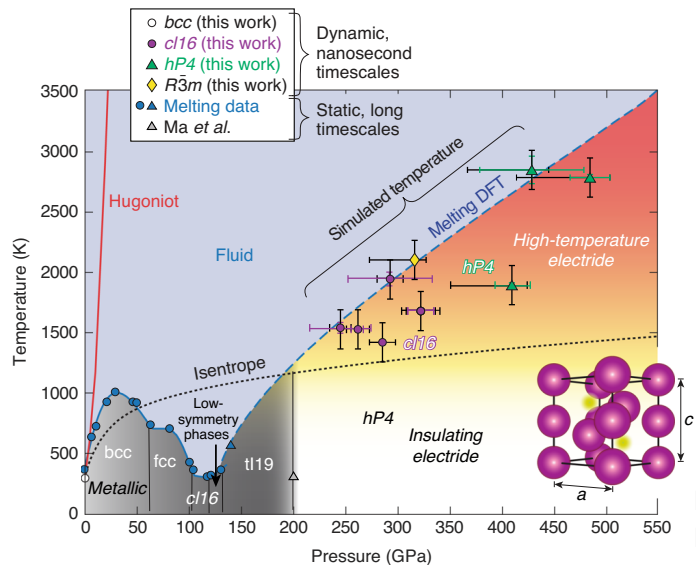


From toy models to materials

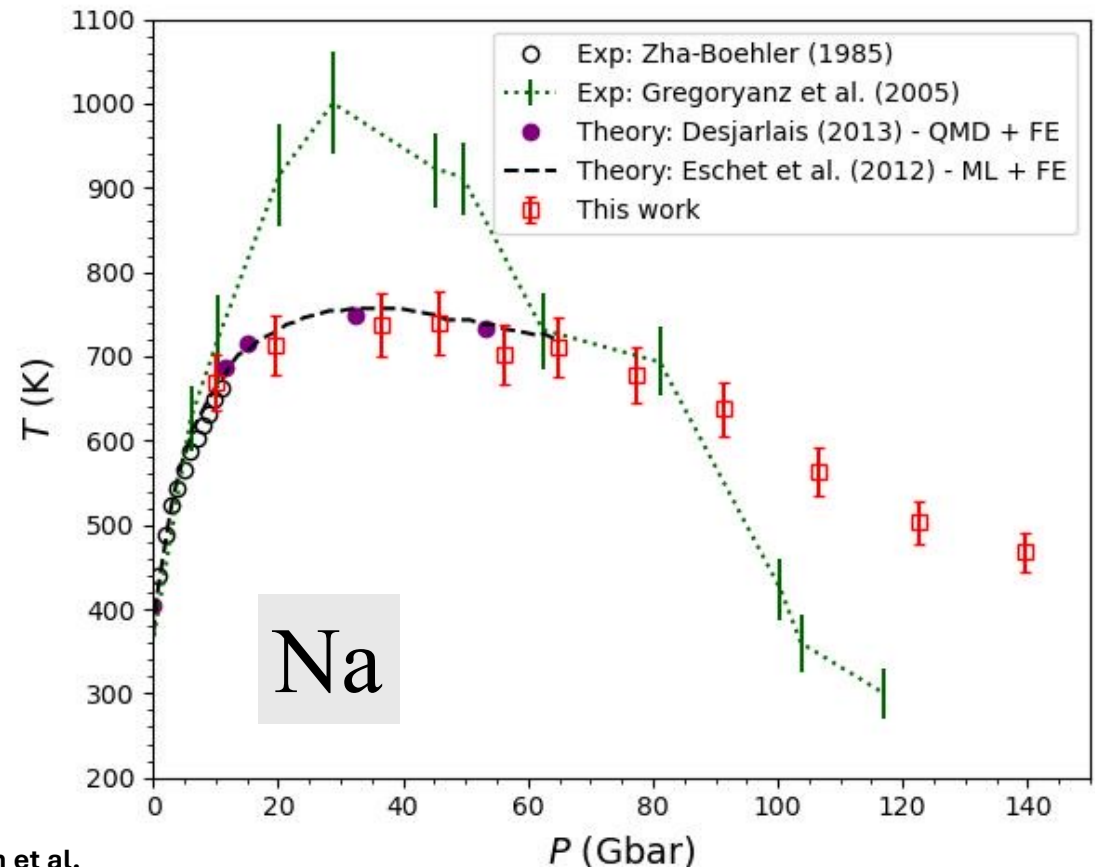


Na: promising results in a tough test

Results from quantum molecular dynamics (VASP) using **GGA** xc and the Na_sv pseudopotential, which treats $2p^6$ and $3s^1$ as valence electrons.

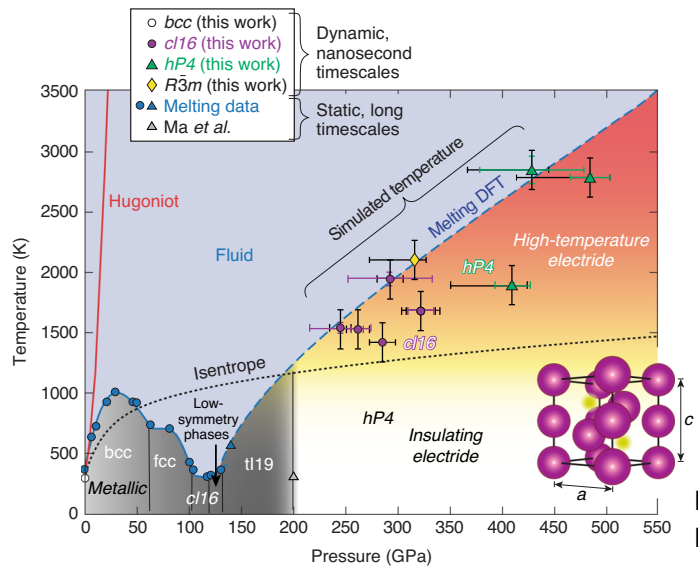


D.N. Poisin et al.
Nat. Comm. (2022)

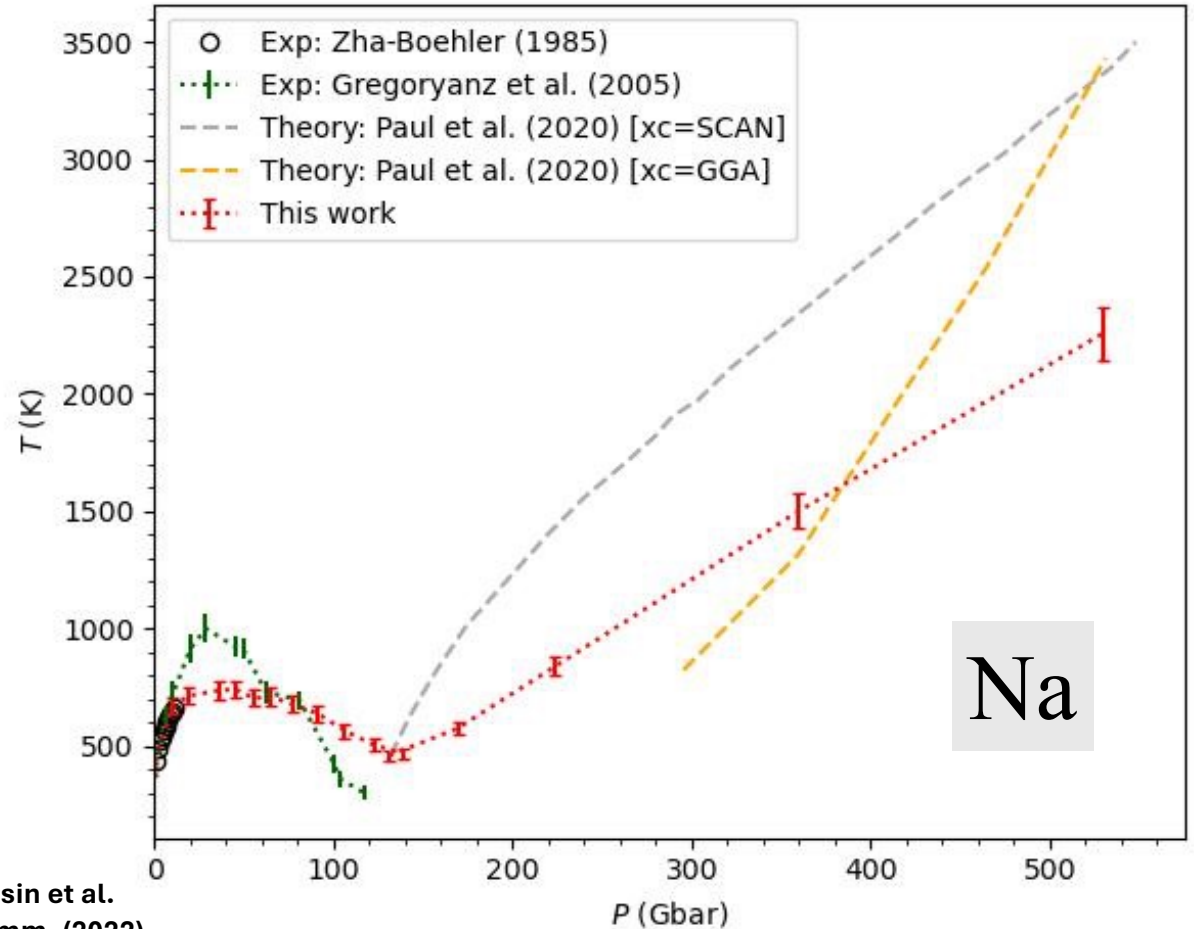


Pushing Na to higher pressures

Results from quantum molecular dynamics (VASP) using **GGA** xc and the Na_{sv} pseudopotential, which treats 2p⁶ and 3s¹ as valence electrons.

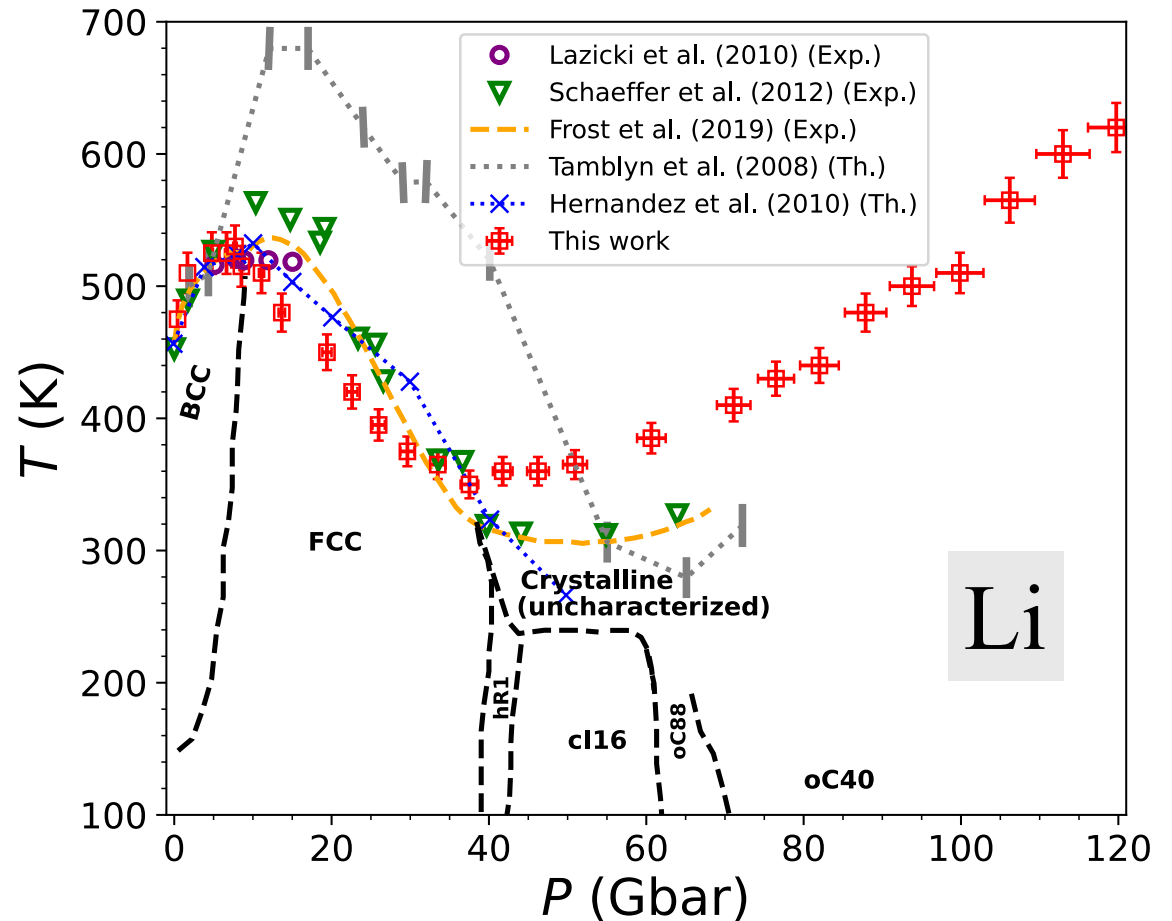


D.N. Poisin et al.
Nat. Comm. (2022)



Li: promising results in another tough test

Results from quantum molecular dynamics (VASP) using **GGA** xc and the Li_sv pseudopotential, which treats $1s^2$ and $2s^1$ as valence electrons

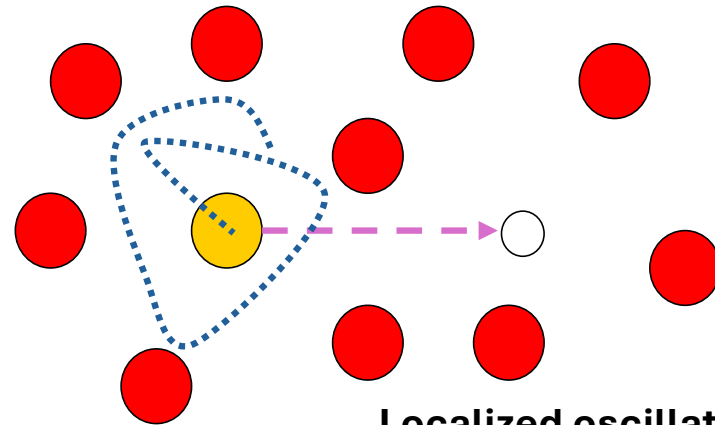
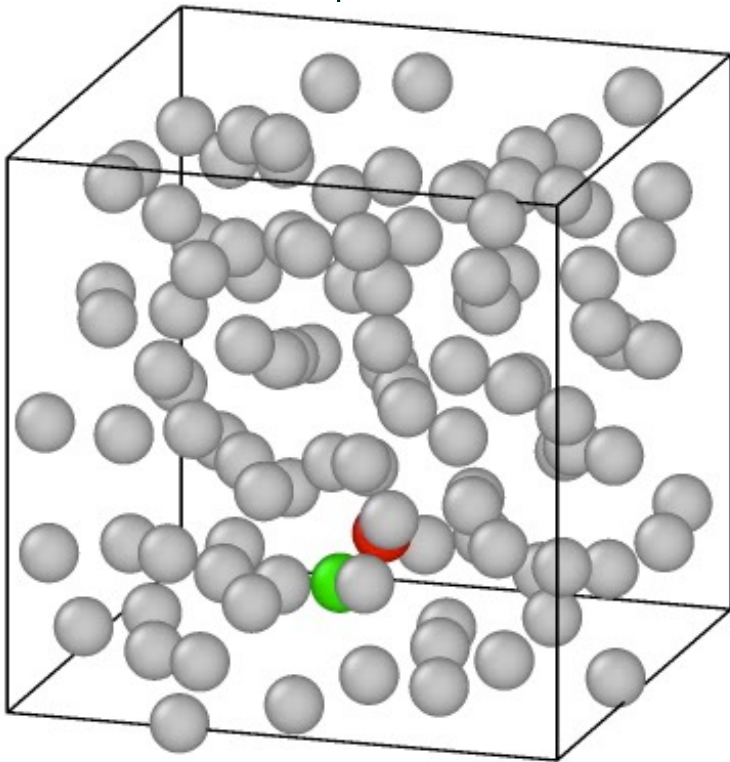


A Minimal Model: formalizing Frenkel's 1926 insight

Über die Wärmebewegung in festen und flüssigen
Körpern.

Von **J. Frenkel** in Leningrad.

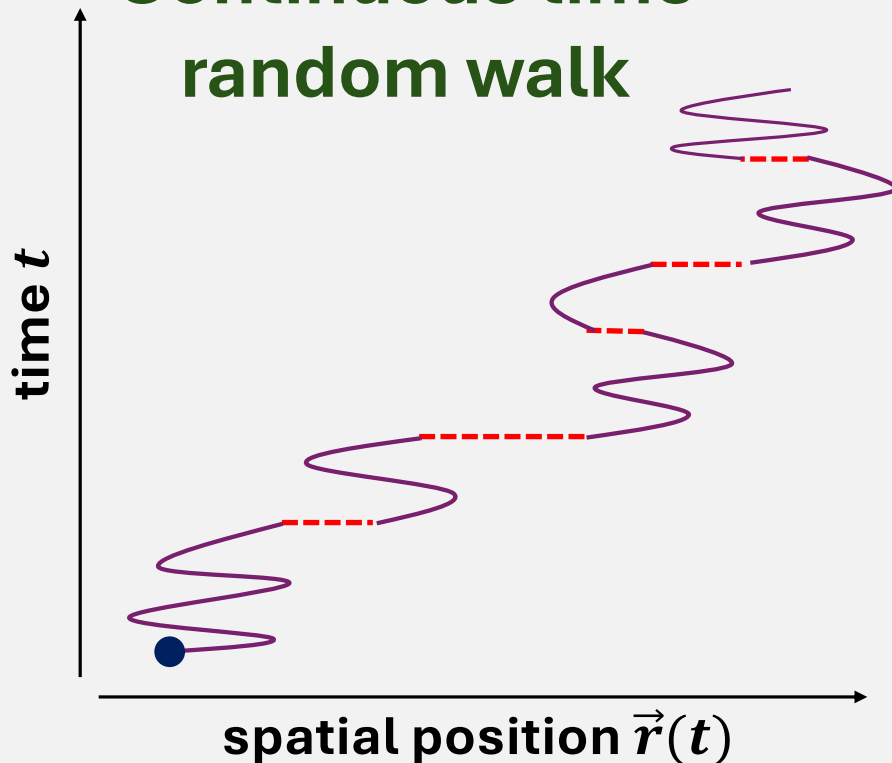
Zeitschrift für Physik 35, 652 (1926)



**Localized oscillations
interrupted by sudden jumps**

A Minimal Model: Formalizing Frenkel's 1926 Insight

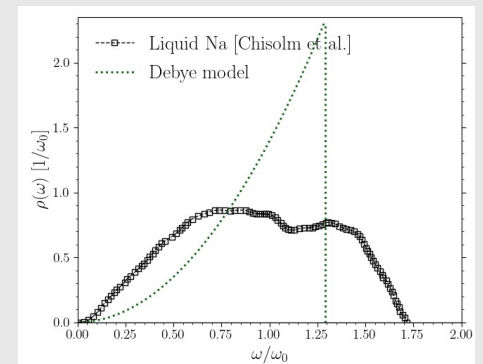
Continuous time random walk



Local harmonic oscillations

Frequency spectrum

$$\rho(\omega)$$



Jumps between local minima

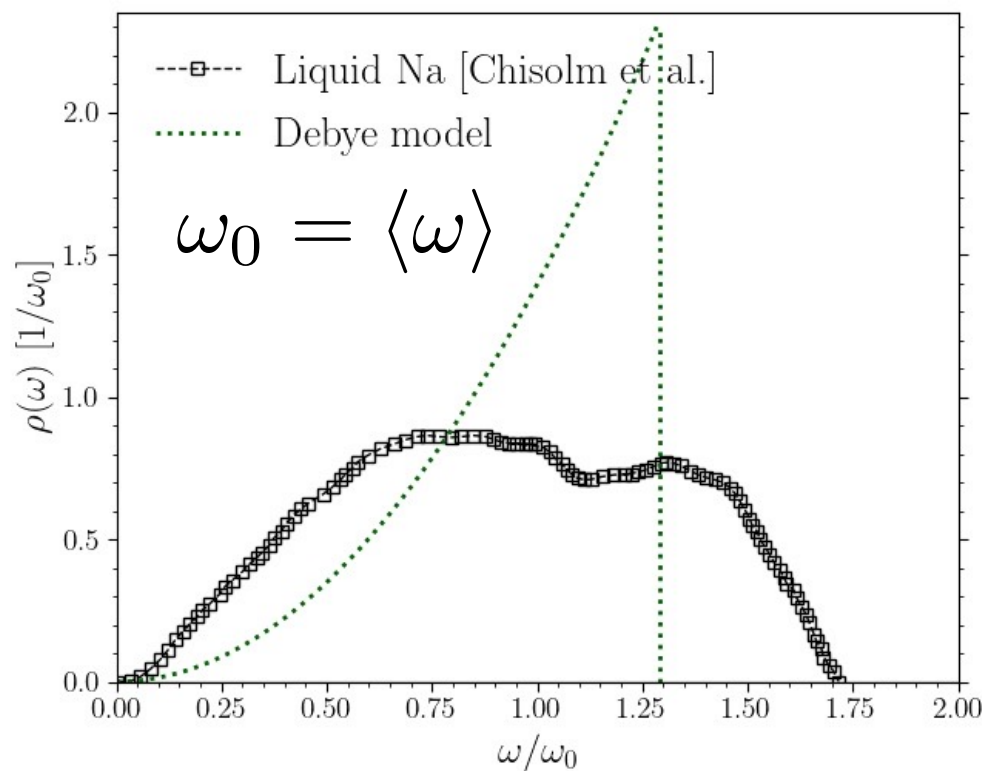
Occurrence of jumps $\phi(t) = \frac{1}{\tau} e^{-t/\tau}$

Jump size $f_J(\mathbf{r}) = \frac{1}{(2\pi l^2)^{3/2}} e^{-r^2/2l^2}$

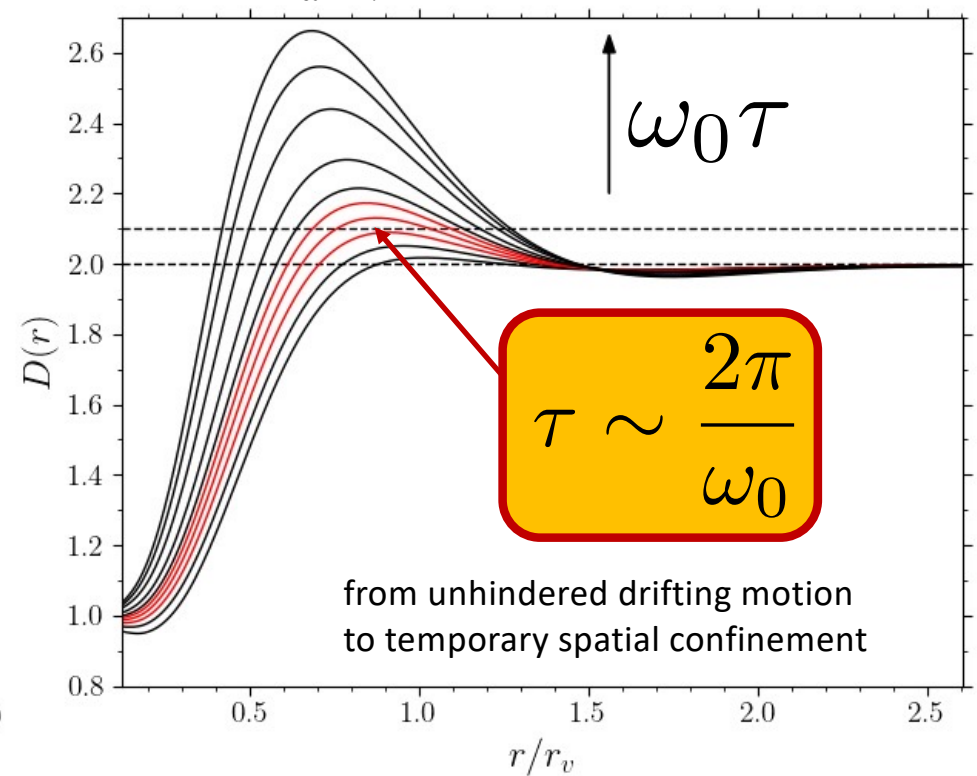
2 independent parameters: $\rho(\omega)$ τ

Emergence of a peak in $D(r)$ near freezing: onset of dynamical hindrance

Frequency spectrum $\rho(\omega)$



$$D(r) = \frac{d \ln \tau(r)}{d \ln r}$$



Key takeaways

- **A signal in the liquid:**

We have identified a robust and reproducible dynamical signature of freezing in elemental liquids, observable in the first-passage time properties of atomic motion.

- **A method without the solid:**

This feature, consistent across interaction potentials, enables melting curves to be determined from liquid-phase simulations alone, without requiring knowledge of the solid phase.

- **A dynamical view of thermodynamic coexistence:**

The threshold value $D(r^*) \approx 2.1$ coincides with the onset of dynamical hindrance in — from unhindered drifting motion to temporary spatial confinement — and offers a new atomistic perspective on the connection to phase coexistence.

Key takeaways

J.D. “*Determining Melting Curves Without Solid-Phase Information*”, under review

A. Farris, R. Perriot and J.D. “*A Liquid-Phase-Only Approach for Efficient Melting Curve Determination via Molecular Dynamics*” , in preparation

An interface for calculating first-passage statistics from MD data will be made publicly available soon

Ongoing and future research directions:

Toward broader applicability and deeper understanding

- **Non-metallic simple liquids and small molecular liquids**

Explore the $D(r)$ criterion in systems like carbon, silicon, and water, where directional or molecular bonding dominates

- **Binary alloys and salts**

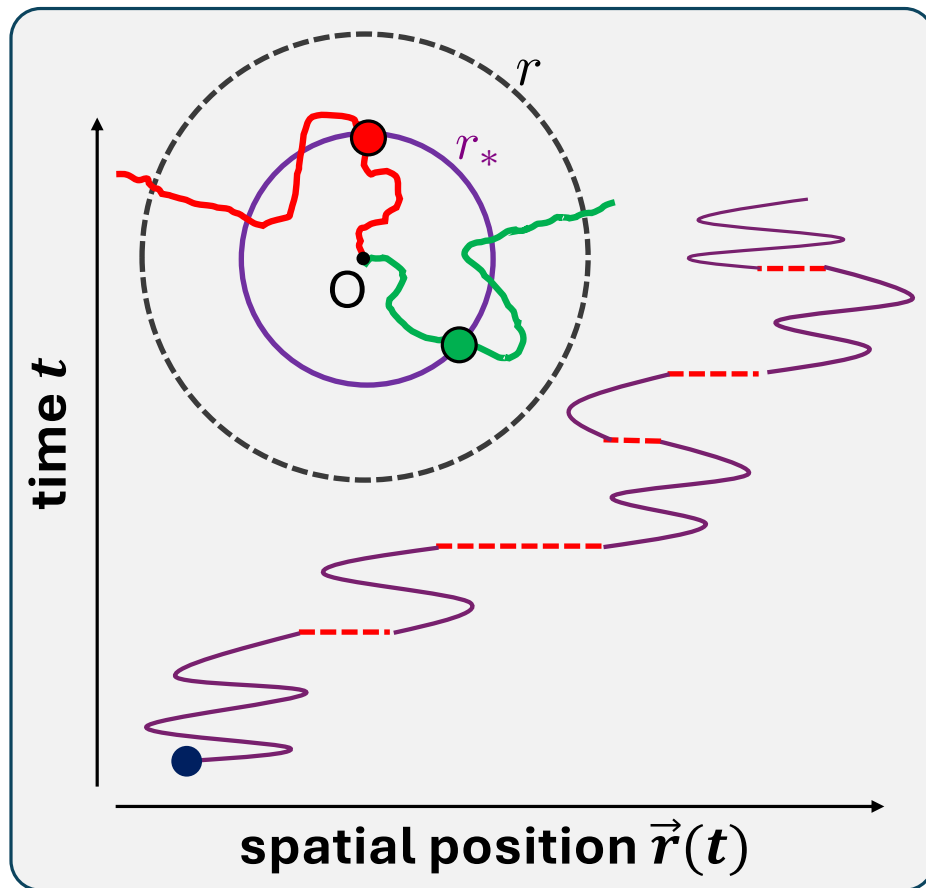
Investigate whether first-passage melting signature persists or shifts under charge or mass asymmetry: Lennard-Jones, Yukawa, NaCl

- **Develop a deeper theoretical understanding**

Formalize the connection between $D(r)$ and thermodynamic coexistence criteria: dual dynamics liquid theory (J.D., J. Stat. Phys., under revision), effect of dimensionality

Backup slides

A Minimal Model: Formalizing Frenkel's 1926 Insight



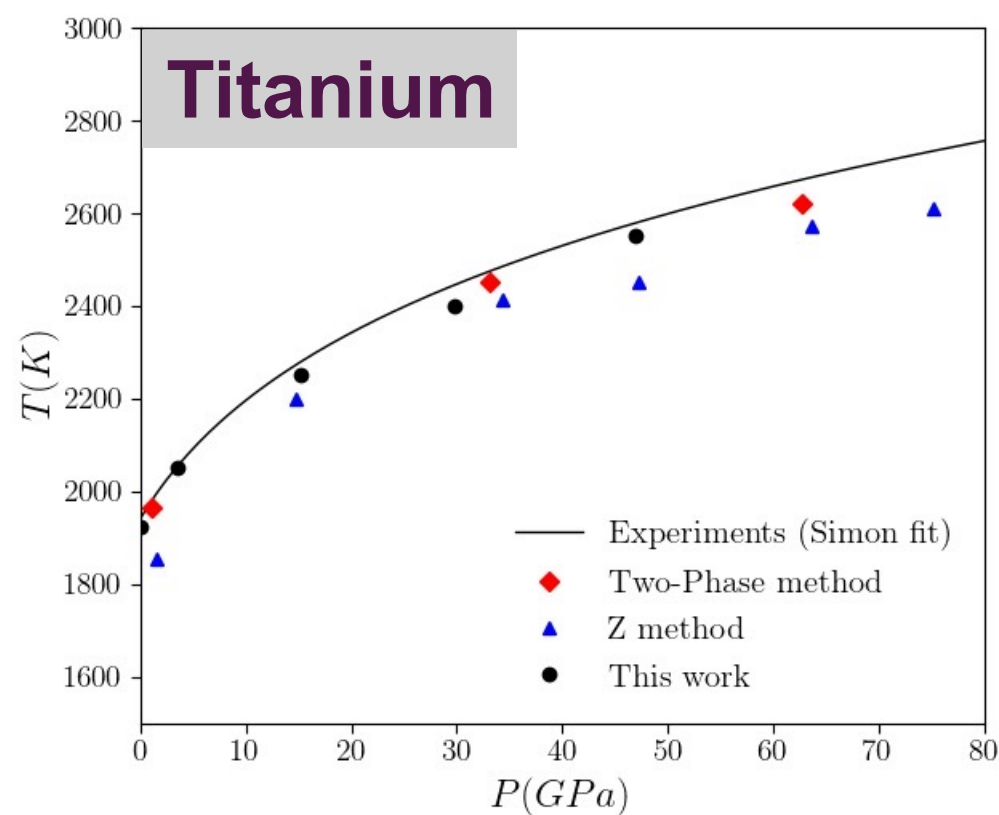
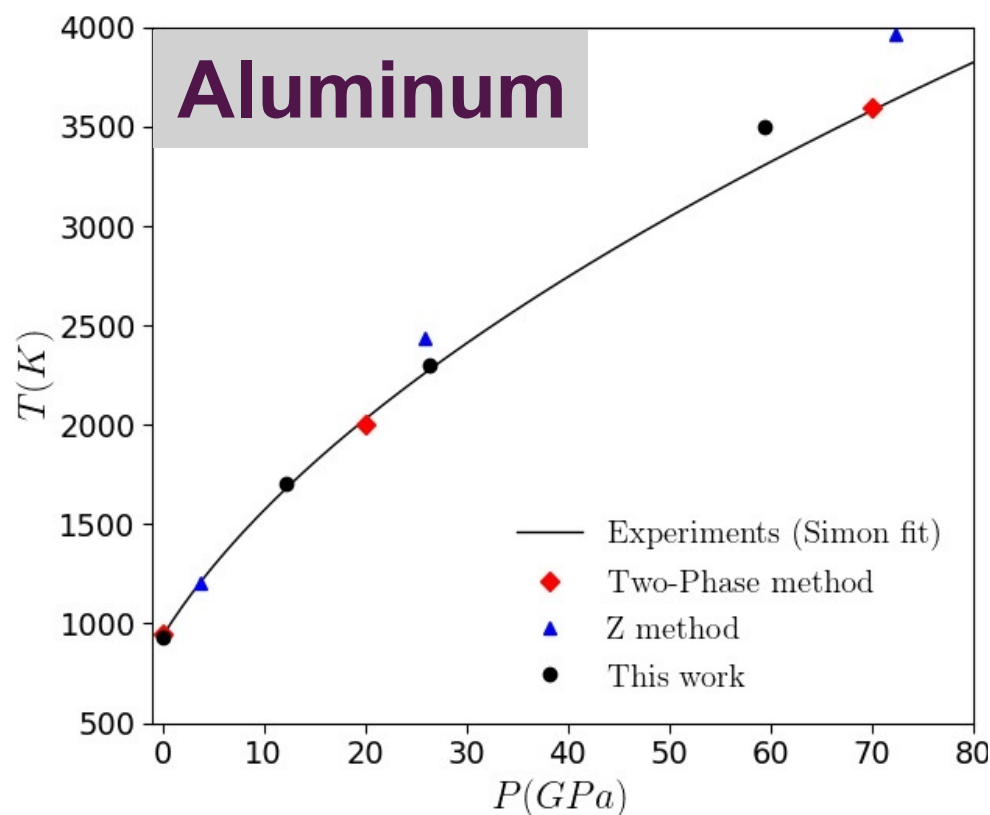
Probability to be at distance r from initial position after a time t

Probability to be at distance r from initial distance r_* after a time $t-t'$

$$\mathcal{P}(r, t|0) = \int_0^t dt' \mathcal{P}(r, t - t'|r_*) F_{r_*}(t')$$

**Probability
of first exit times**

From toy models to materials



Revealing the Physics Beneath the Laws



“Without prejudice to its unshakable importance, thermodynamics requires the cultivation of mechanical models representing it in order to deepen our knowledge of nature – not in spite of, but rather precisely because these models do not always cover the same ground as thermodynamics, but instead offer a glimpse of a new viewpoint.”

Ludwig Boltzmann
Lectures on Gas Theory (1896)

“Thermodynamical results are generally highly accurate. On the other hand, it is sometimes rather unsatisfactory to obtain results without being able to see in detail how things really work, so that it is very often convenient to complete a thermodynamical result with at least a rough kinetic interpretation.”

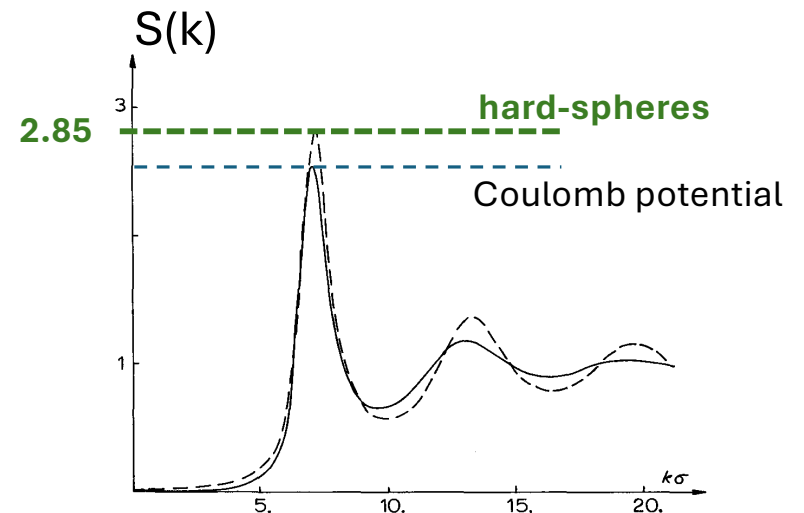
Thermodynamics (1936)



Can We Do Without the Solid?

Method	Requires solid phase	Comment
Free energy	Yes	Exact in principle, expensive in practice
Two phase coexistence	Yes	Physically grounded, setup is delicate simulations are demanding
Lindemann	Yes	Empirical threshold on atomic displacements
Z method	Yes	Based on superheating the solid
Heat-until-melt	Yes	Superheating limit, gives upper bound

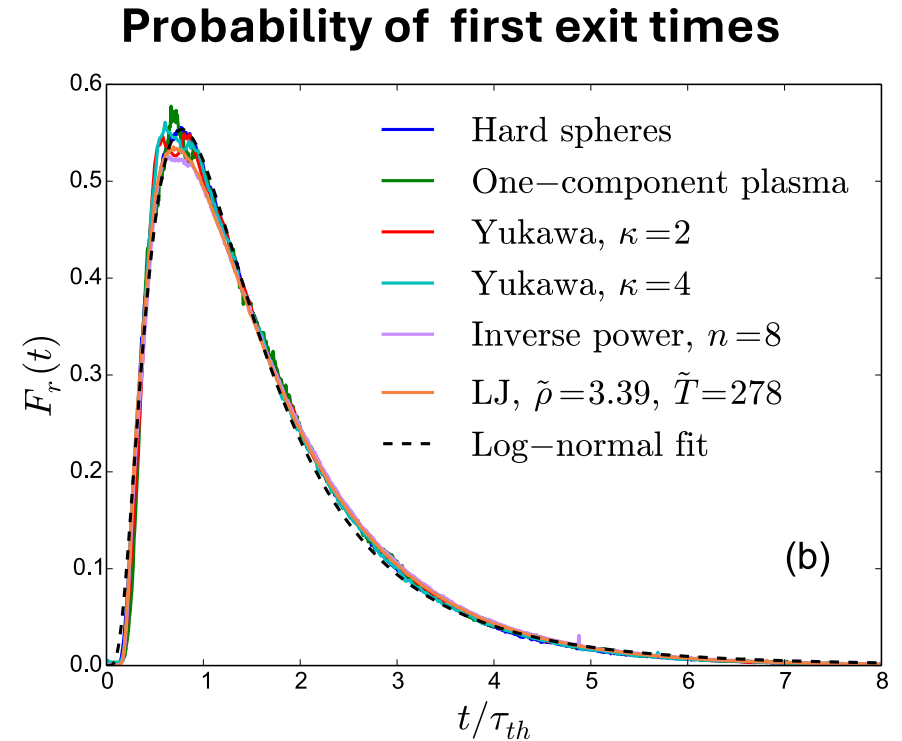
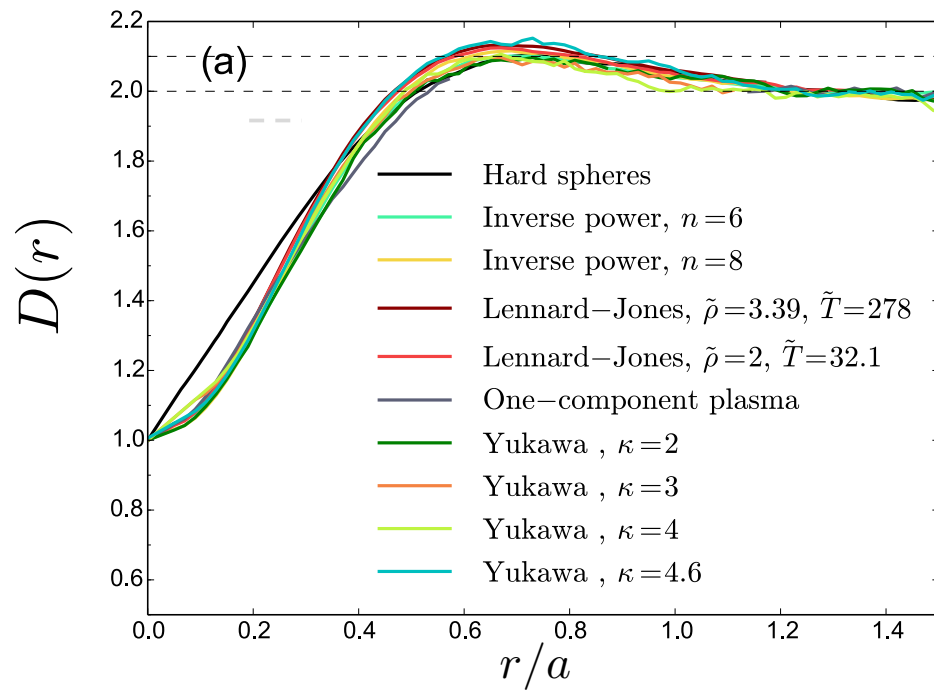
Hansen-Verlet No **Heuristic, not widely transferable ($\pm 15\%$)**

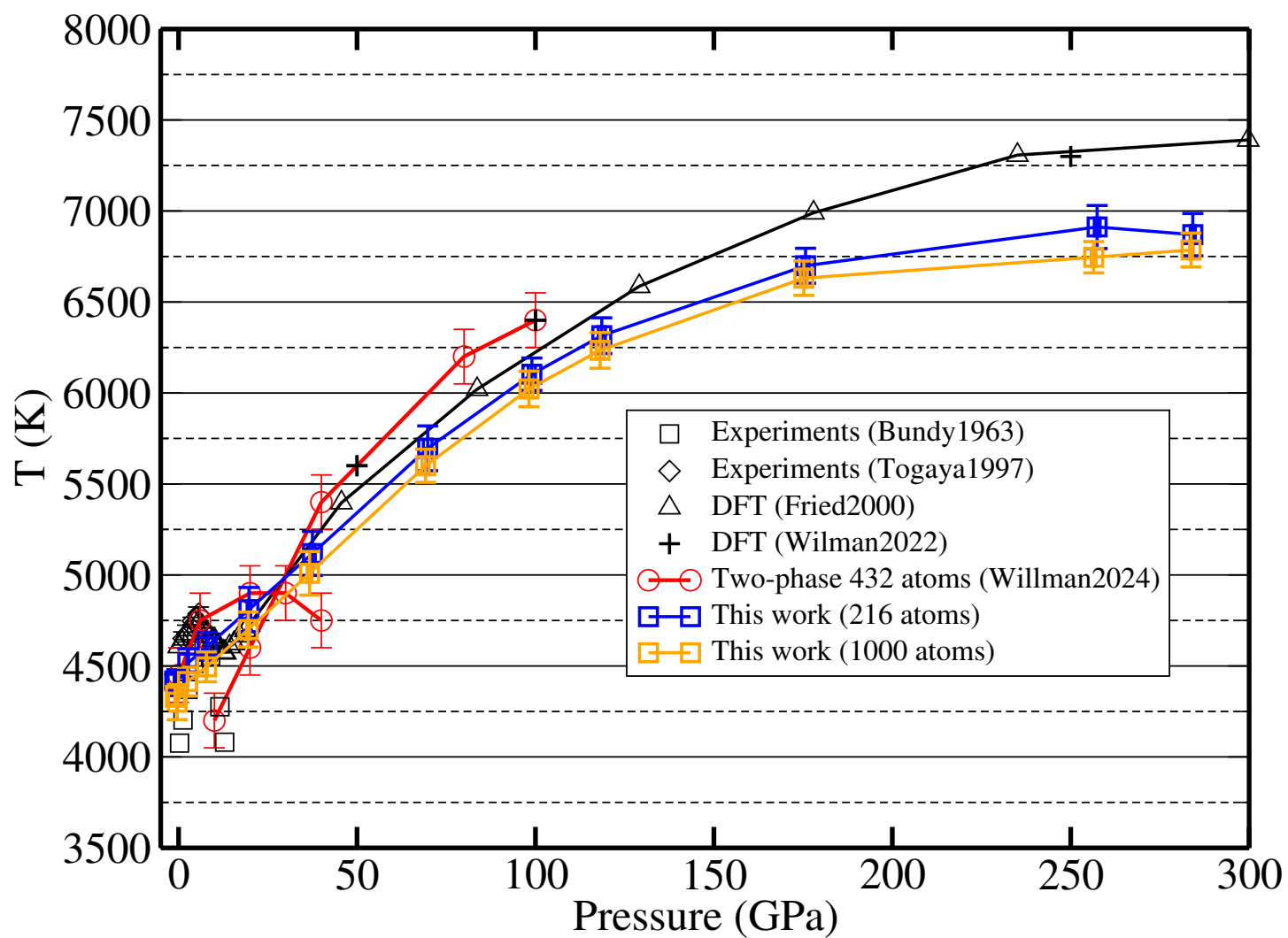


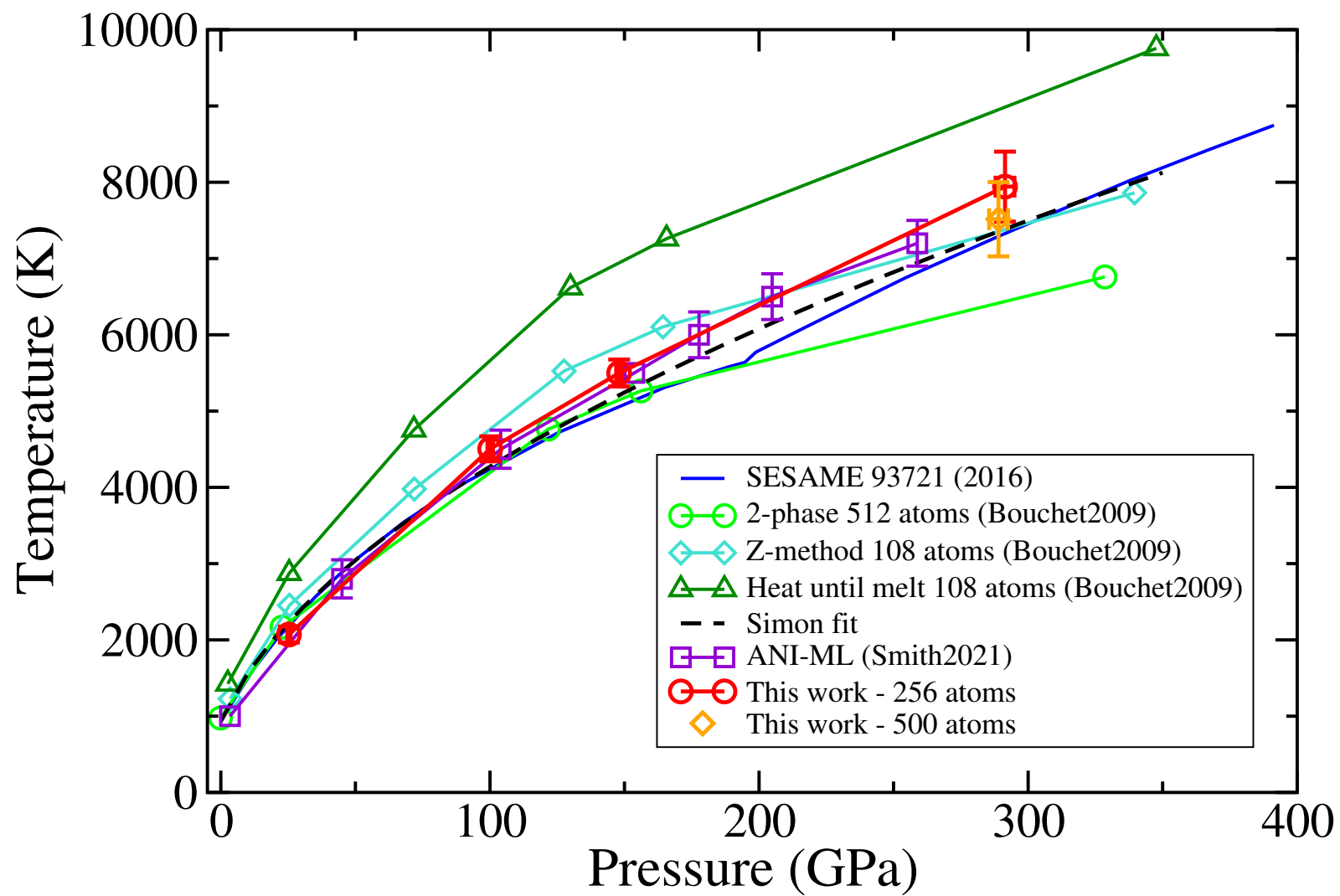
Hansen et al., Mol. Phys. 25, 1281 (1973)

Hansen-Verlet criterion:
a liquid freezes when the first peak
of its *static structure factor* $S(k)$
reaches $S(k)_{\max} = 2.85$

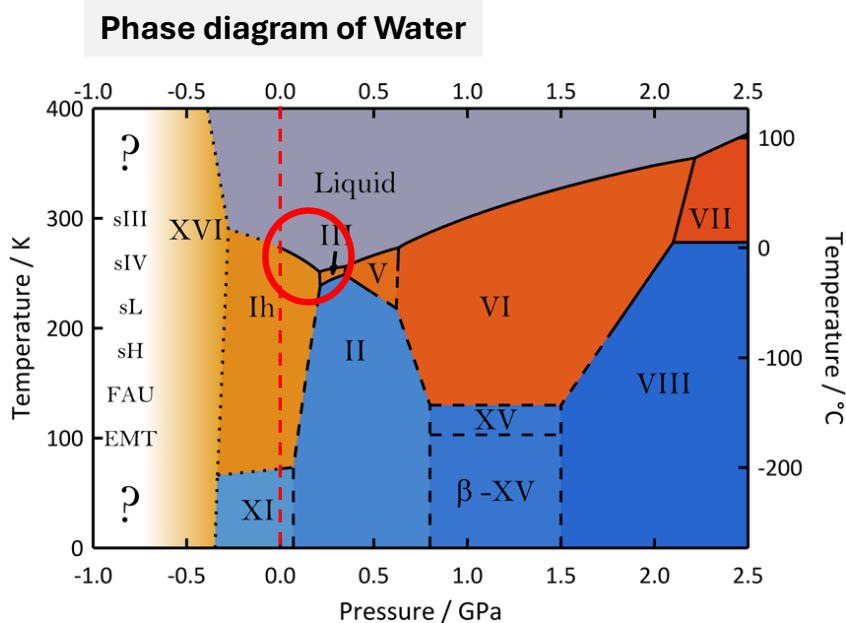
The consistency extends beyond the average property







Melting Curves: Revealing Physics Then and Now



THEORETICAL CONSIDERATIONS ON THE EFFECT OF PRESSURE
IN LOWERING THE FREEZING POINT OF WATER.

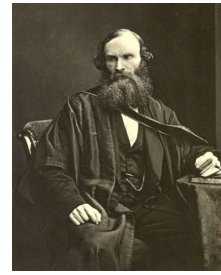
By JAMES THOMSON,

*(Taken, with some slight alterations made by the author, from the Transactions
of the Royal Society of Edinburgh, vol. xvi. part v. 1849.)*

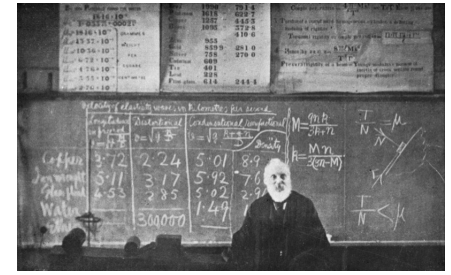
SOME time ago my brother (Professor William Thomson) pointed out to me a curious conclusion to which he had been led, by reasoning on principles similar to those developed by Carnot, with reference to the motive power of heat. It was, that *water at the freezing point may be converted into ice by a process solely mechanical, and yet without the final expenditure of any mechanical work.* This at first appeared to me to involve an impossibility, because water expands while freezing; and therefore it seemed to follow, that if a quantity of it were merely enclosed in a vessel with a moveable piston and frozen, the motion of the piston, consequent on the expansion, being resisted by pressure, mechanical work would

The Cambridge and Dublin mathematical journal
(1849)

James Thomson



William Thomson (Lord Kelvin)



XII. *The Effect of Pressure in Lowering the Freezing-Point of Water experimentally demonstrated.* By Professor W. THOMSON, Glasgow*.

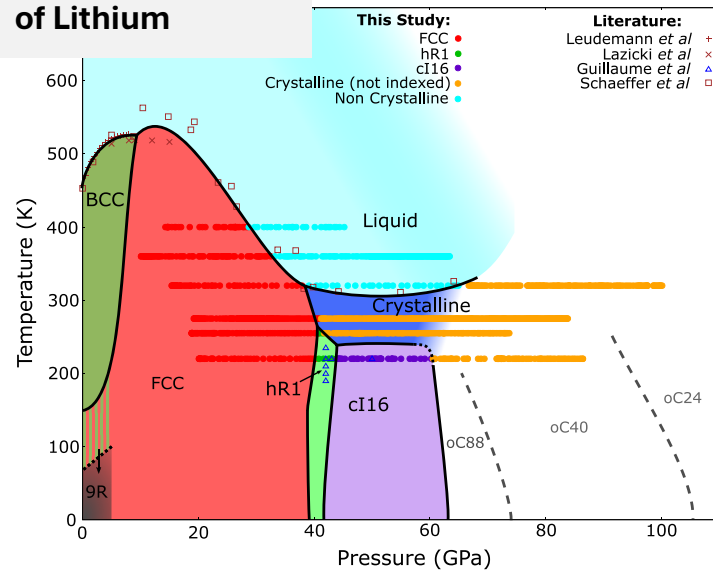
ON the 2nd of January 1849, a communication entitled "Theoretical Considerations on the Effect of Pressure in Lowering the Freezing-Point of Water, by James Thomson, Esq., of Glasgow," was laid before the Royal Society, and it has since been published in the Transactions, vol. xvi. part 5†. In that paper it was demonstrated that, if the fundamental axiom of Carnot's Theory of the Motive Power of Heat be admitted, it follows, as a rigorous consequence, that the temperature at which ice melts will be lowered by the application of pressure; and the extent of this effect due to a given amount of pressure was deduced by a reasoning analogous to that of Carnot from Regnault's experimental determination of the latent heat, and the pressure of saturated aqueous vapour at various temperatures differing very little from the ordinary freezing-point of water. Reducing to Fahrenheit's scale the

The London, Edinburgh, and Dublin Philosophical Magazine **(1850)**

Melting Curves: Revealing Physics Then and Now

Phase diagram
of Lithium

Frost et al. (2019)

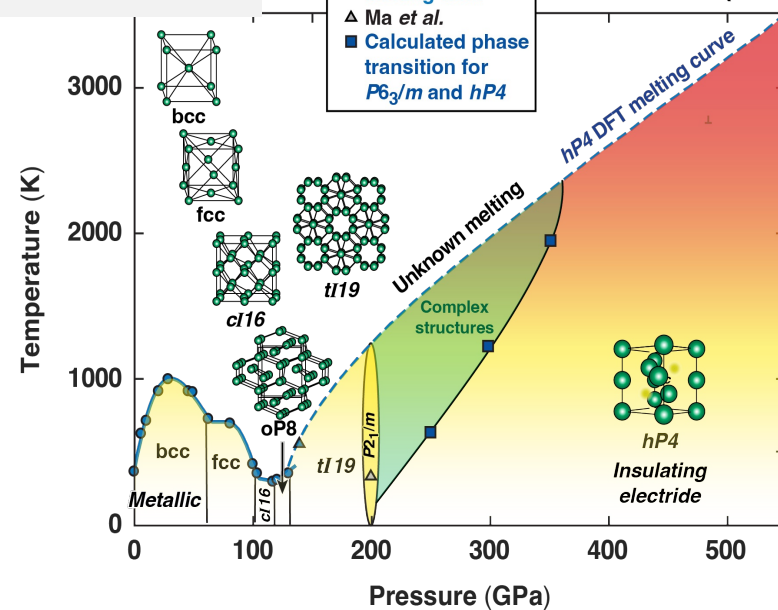


Melting curves remain difficult to determine experimentally.
Theoretical prediction is essential.

Elemental materials still catch us off guard —
melting curves reflect the evolving nature of
bonding under pressure.

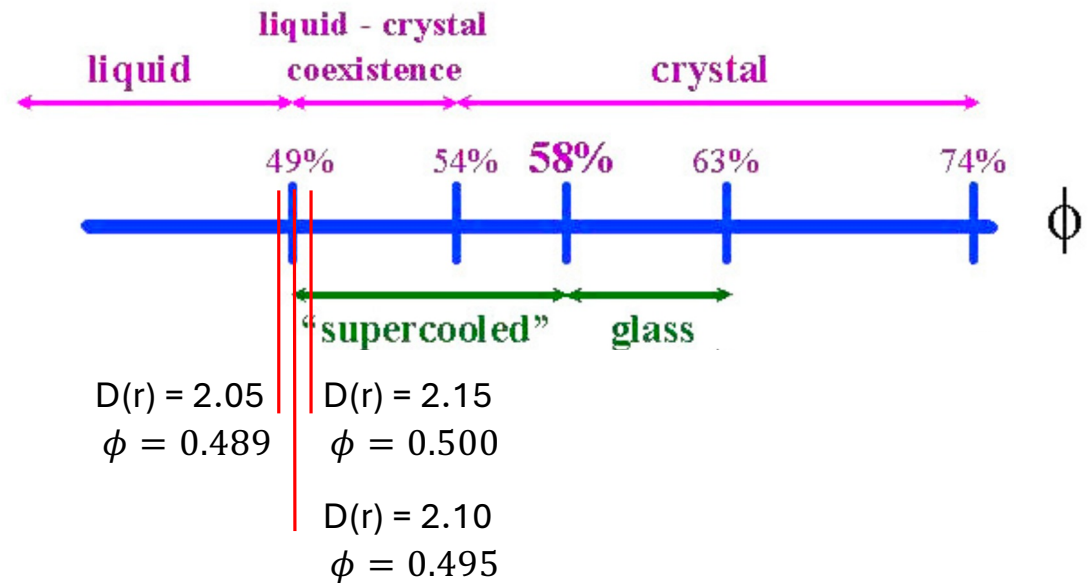
Phase diagram
of Sodium

Polsin et al. (2022)



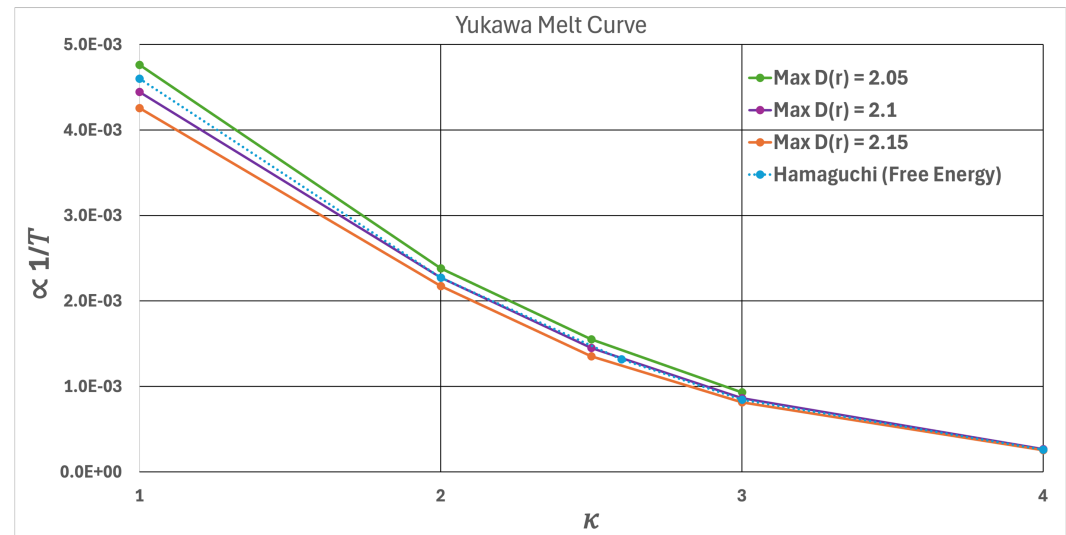
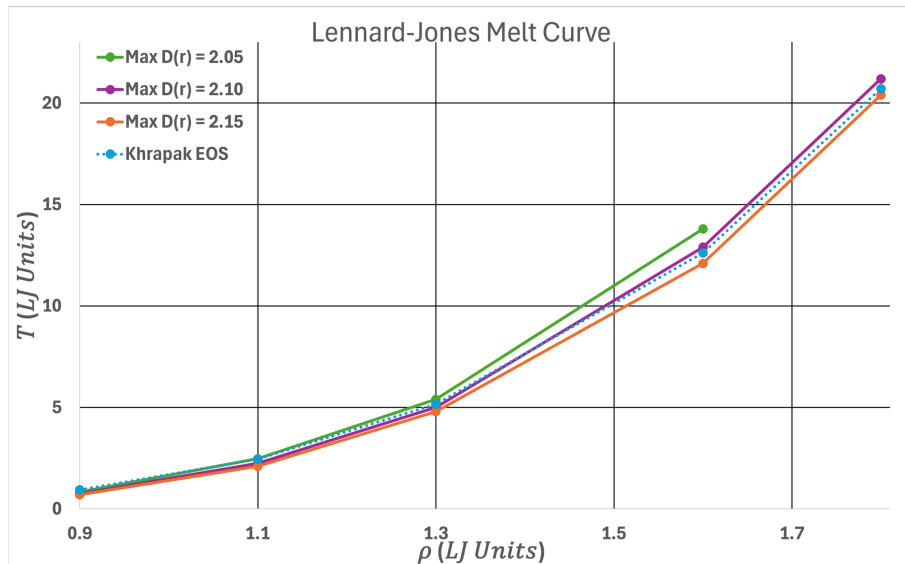
We determined that $D(r) = 2.1$ corresponds to the melting point for hard spheres

- Accurate to within 1% ($\phi_{exact} = 0.4918$)



Pieprzyk et. al., 2019
V. Prasad. et. al., 2007

We determined that $D(r) = 2.1$ corresponds to the melting point for Lennard-Jones and Yukawa systems



Khrapak et. al., 2010
Hamaguchi et. al., 1997

Can we develop an efficient to compute melting curves using $D(r)$?

- **Statistical errors due to different MD trajectories**
 - **Systematic Errors:**
 - Size effects (model dependent)
 - Subsampling ($N = 1, 16, 32, \dots$)
 - Simulation length
 - Maximum value of $D(r)$
 - Asymptotic behavior in tail of $D(r)$
 - Interaction Potential
 - Truncation and smoothing at cutoff
 - Range of interactions
 - Thermodynamic ensemble in MD simulation
 - Classical MD (NVT, NPT, NVE)
 - Quantum MD
 - MD Control Parameters
 - Thermostat parameters (appropriate fluctuations)
- **What is the optimal value of $D(r)$ at melting?**
- **How does our method compare against other MD methods?**